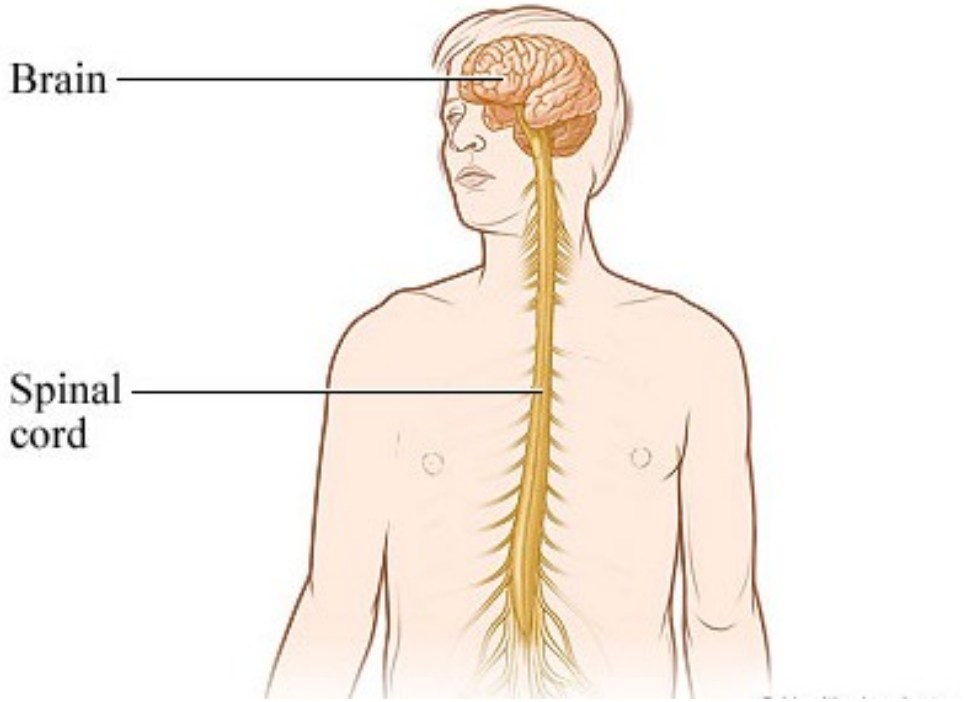
The Central Nervous System (CNS) - protection



Central Nervous System

- Brain and Spinal Cord
- Interprets, Coordinates and Integrates
  - input from sensory nervous system
  - output to motor nervous system

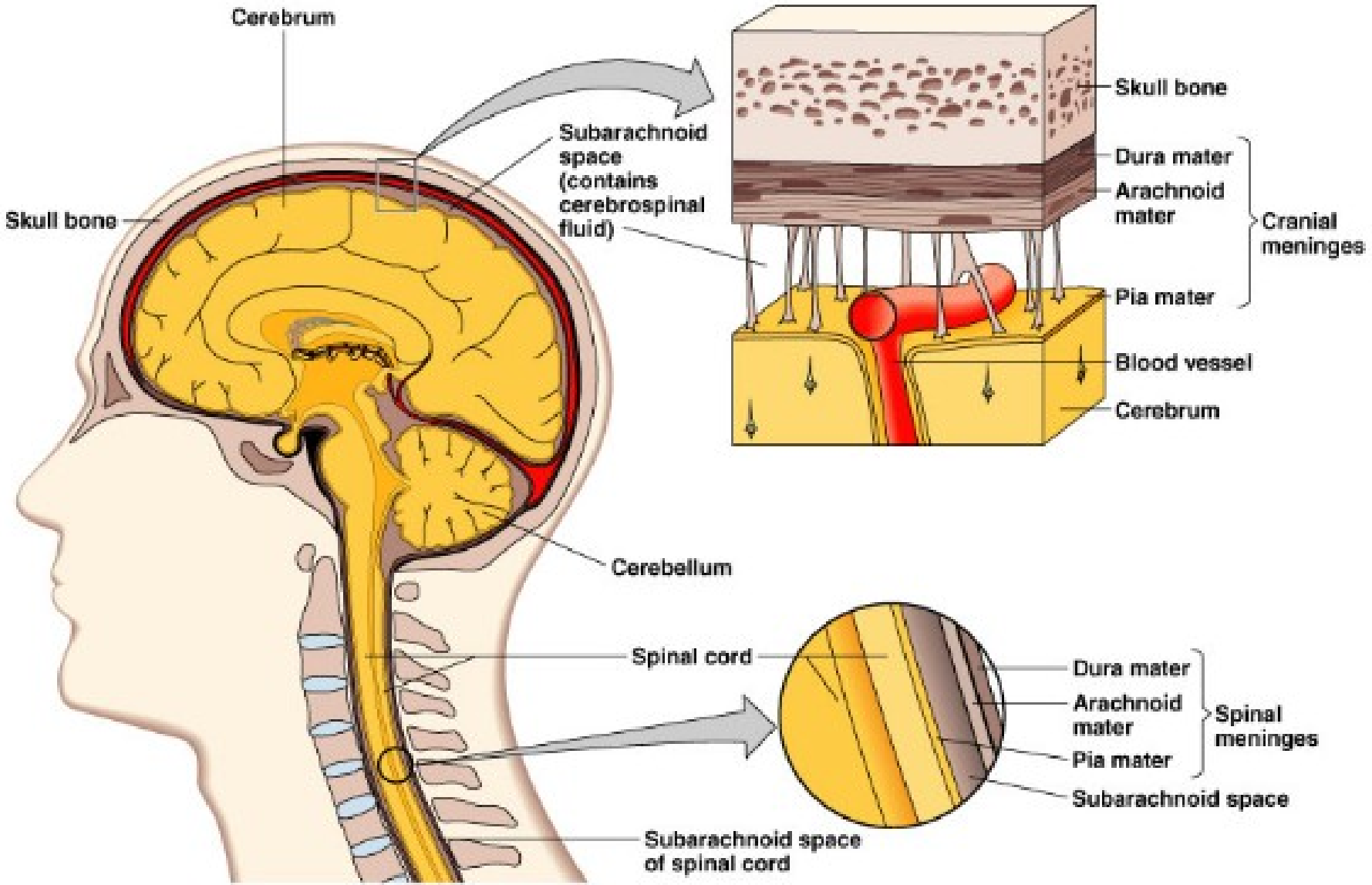
Protection of CNS

- Delicate and Vital
- Fragile, does not heal or regenerate well
- Heavily Protected
- 3 structures for protection
  - Bone
  - Meninges
  - Cerebro-Spinal Fluid (CSF)

Protection of CNS

- **Bone**
  - Cranium (skull) houses the brain
  - Spinal cord within vertebral canal in vertebrae

- **Meninges**
  - 3 layers of connective tissue
  - Between bone and CNS
  - Fibrous outer layer
  - Loose mesh middle layer
  - Delicate inner layer
    - Blood vessels
    - Sticks closely to surface of brain, and spinal cord

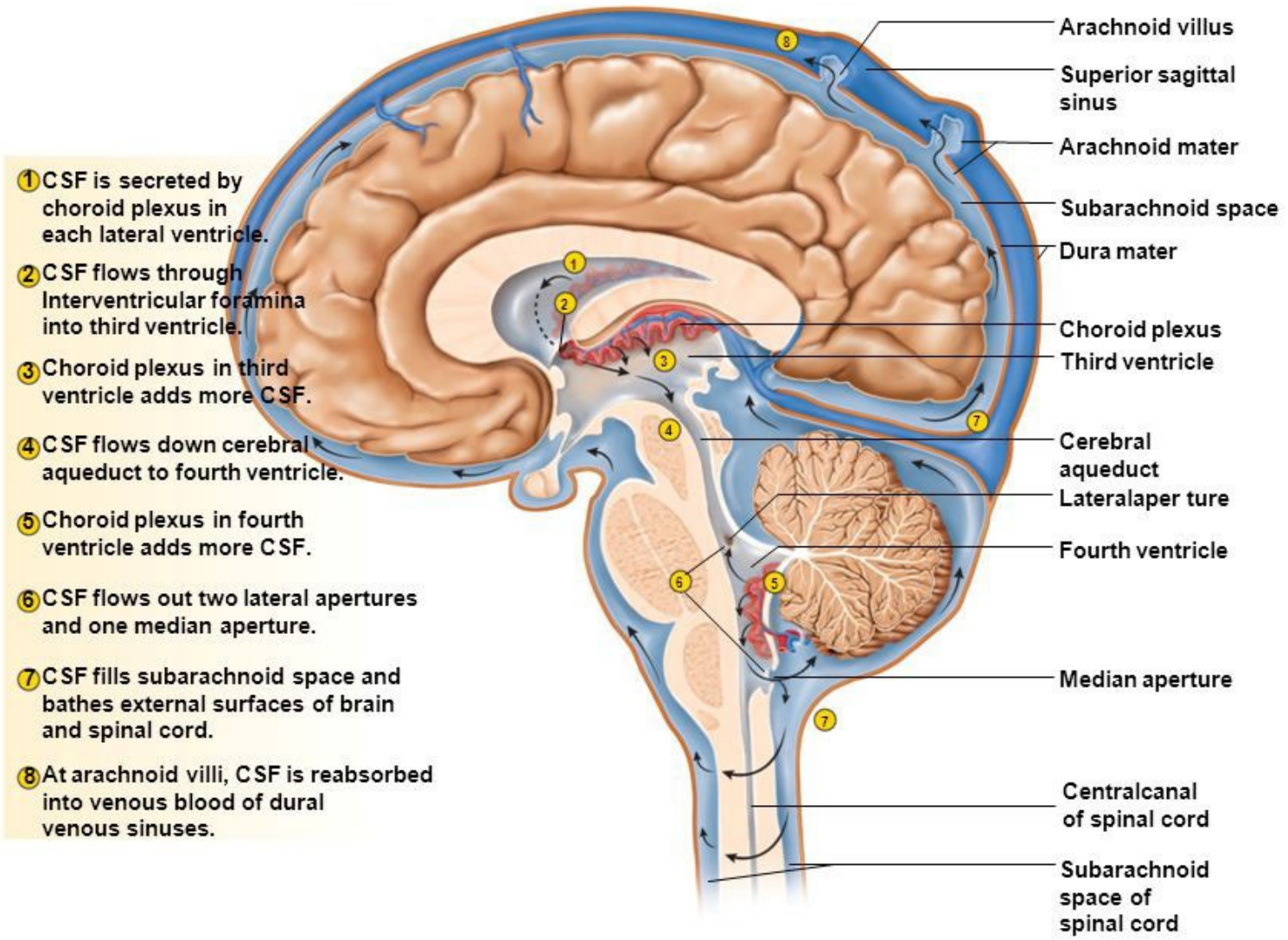

Cerebrum
Skull bone
Subarachnoid space (contains cerebrospinal fluid)
Cerebellum
Spinal cord
Subarachnoid space of spinal cord
Skull bone
Dura mater
Arachnoid mater
Pia mater
Cranial meninges
Blood vessel
Cerebrum
Dura mater
Arachnoid mater
Pia mater
Subarachnoid space
Spinal meninges

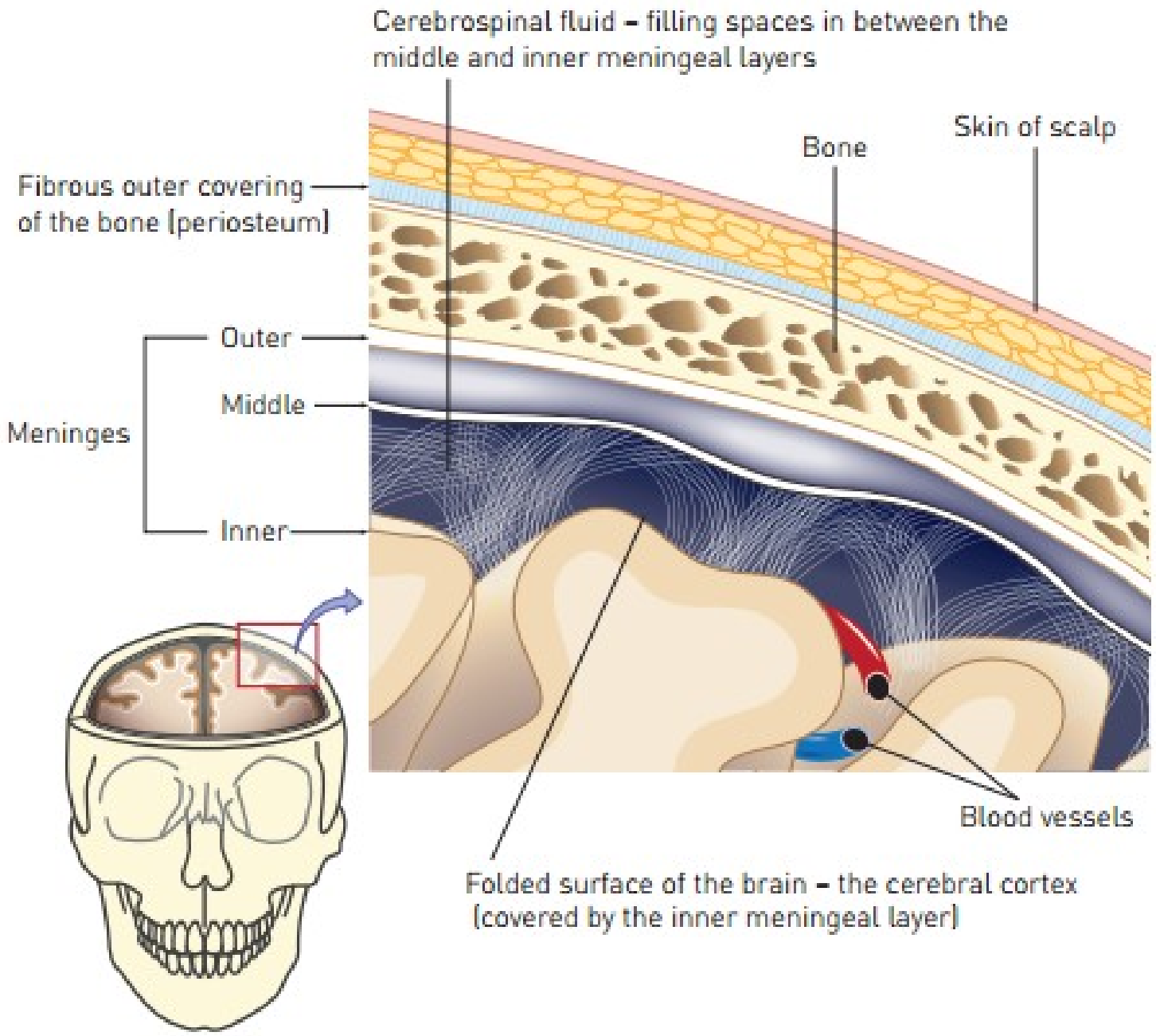
Protection of CNS

- **Cerebrospinal Fluid (CSF)**
  - 3 functions:
    - Protection
    - Support
    - Transport
  - Between middle and inner layers of meninges
  - Circulates through cavities in brain and canal in spinal cord
  - Clear, watery fluid:
    - Few cells
    - Glucose, protein, urea, salts
  - Shock absorber, supports fragile brain tissue
  - Formed from blood, circulates and eventually re-enters blood.
  - Barrier between blood vessels and CSF is quite tight
    - Protects against entry of bacteria, and some other particles
  - Transports nutrients and wastes to cells of brain and spinal cord

Information Only: This is more detail than you need to know.

Flow of Cerebrospinal Fluid




Cerebrospinal fluid – filling spaces in between the middle and inner meningeal layers
Skin of scalp
Bone
Fibrous outer covering of the bone (periosteum)
Outer
Middle
Meninges
Inner
Blood vessels
Folded surface of the brain – the cerebral cortex (covered by the inner meningeal layer)

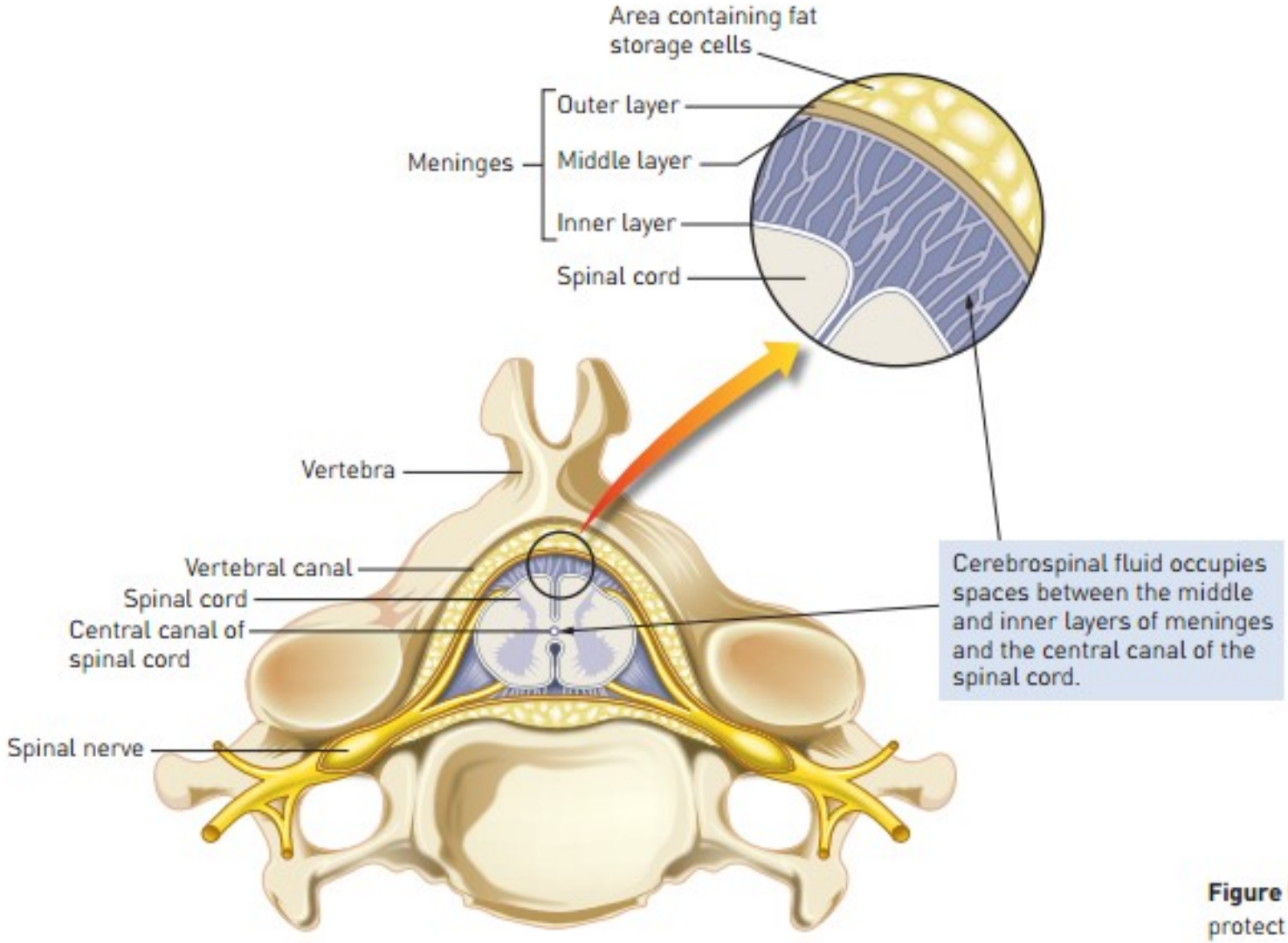


Figure 5.3 Structures that protect the spinal cord